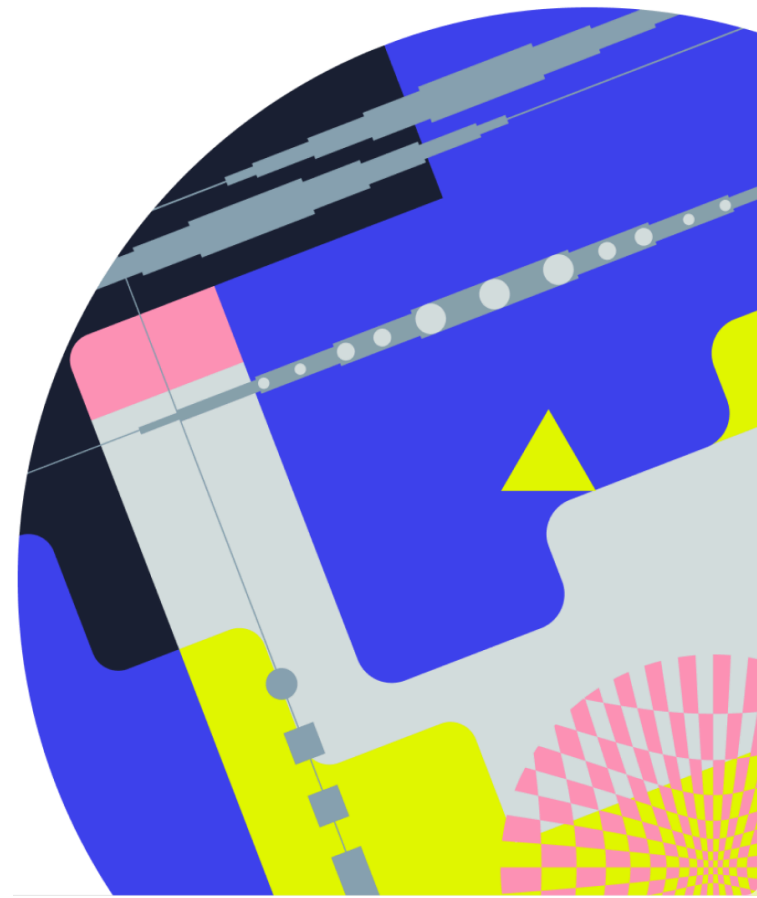
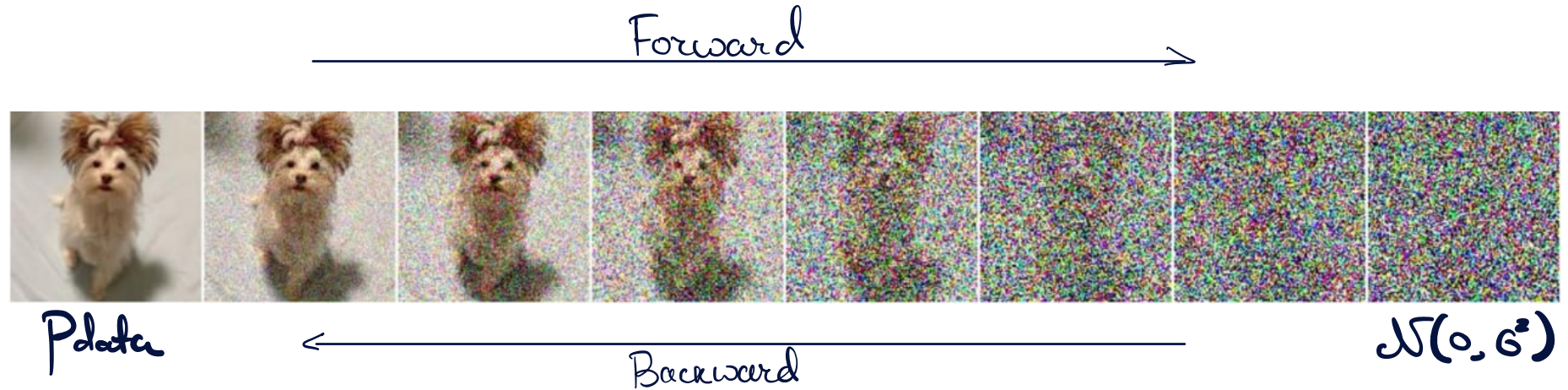


Continuous-time diffusion models

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Diffusion



Idea: define forward noising process and try to reverse it.

Discrete time

- Variance preserving (VP) process: $t=1, \dots, T$

$$X_{t+1} = \sqrt{1-\beta_t} X_t + \sqrt{\beta_t} \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, I)$$

$$(If \text{ Var } X_0 = I \Rightarrow \text{Var } X_t = I)$$

$$p_{t|0}(x_t | x_0) = \mathcal{N}(x_t | \alpha_t \cdot x_0, \sigma_t^2 I) \quad \begin{array}{l} \alpha_t \rightarrow 0 \\ \sigma_t^2 \rightarrow 1 \end{array}$$

$$p_T(x_T) \approx \mathcal{N}(x_T | 0, I)$$

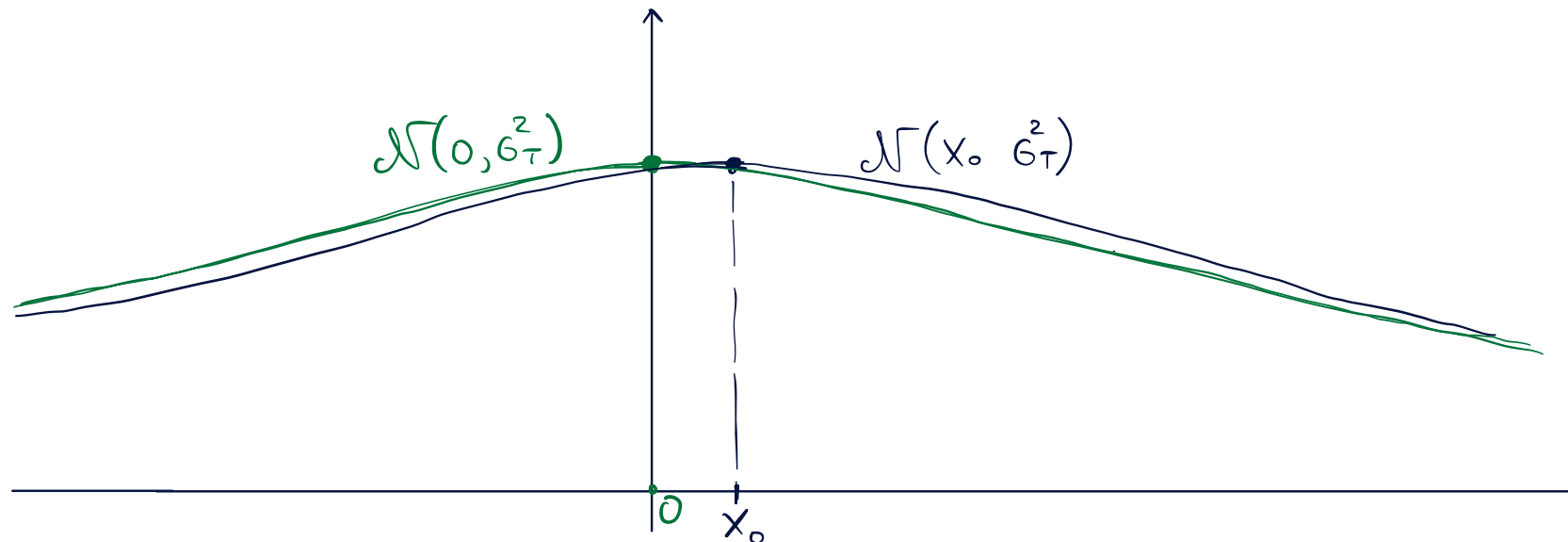
Discrete time

- Variance exploding (VE) process: $t=1, \dots, T$

$$X_{t+1} = X_t + g(t) \cdot \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, I)$$

$$P_{t|0}(x_t | x_0) = \mathcal{N}(x_t | x_0, G_t^2 I), \quad G_t^2 = \sum_{s \leq t} g^2(s)$$

If G_T^2 is big $\Rightarrow P_T(x_T) \approx \mathcal{N}(0, G_T^2 I)$



ODEs

$$dX_t = f(X_t, t)dt \quad / \quad \dot{X} = f(X, t)$$

Euler scheme: $X_{t+h} \approx X_t + h \cdot f(X_t, t)$

• Part of VP process: $X_{t+1} = \sqrt{1 - \beta_t'} X_t + \sqrt{\beta_t'} \epsilon_t$

• Cont. modification:

$$X_{t+h} = \sqrt{1 - \beta_t' h} X_t = \left(1 - h \frac{\beta_t}{2} + o(h)\right) X_t$$

$$\frac{X_{t+h} - X_t}{h} = \frac{-h \frac{\beta_t}{2} + o(h)}{h} \cdot X_t$$

$\downarrow$
 $\dot{X}_t$

$\downarrow$
 $-\frac{\beta_t}{2} X_t$

$$\text{Solution: } X_t = X_0 \cdot \exp\left(-\frac{1}{2} \int_0^t \beta_s ds\right)$$

Wiener process

VE process with $g(t) = 1$:

$$X_{t+1} = X_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, I) \text{ ind. with } X_t.$$

- Let $X_0 = 0$
- Independent increments:
 $t_1 < \dots < t_m \Rightarrow X_{t_1}, X_{t_2} - X_{t_1}, \dots, X_{t_m} - X_{t_{m-1}}$ are indep.
- $X_t - X_s = \sum_{i=s+1}^t \varepsilon_i \sim \mathcal{N}(0, (t-s)I).$

Def Wiener process is a continuous-time process

- $W_0 = 0$
- It has indep. increments
- $W_t - W_s \sim \mathcal{N}(0, (t-s)I)$
- It has continuous trajectories

SDEs

- VE process: $X_{t+1} = X_t + g(t) \varepsilon_t$
- VP process: $X_{t+1} = \sqrt{1 - \beta_t'} X_t + \sqrt{\beta_t'} \varepsilon_t$

Continuous limits?

Def $dX_t = f(X_t, t)dt + G(X_t, t)dW_t$, if
$$X_{t+h} \approx X_t + f(X_t, t) \cdot h + \underbrace{G(X_t, t)(W_{t+h} - W_t)}_{\text{indep. with } X_t}$$

Euler scheme:

$$X_{t+h} \approx X_t + f(X_t, t) \cdot h + G(X_t, t) \cdot \sqrt{h'} \cdot \varepsilon_t,$$
$$\varepsilon_t \sim \mathcal{N}(0, I) - \text{indep. with } X_t$$

SDEs

Discrete VE: $X_{t+1} = X_t + g(t) \varepsilon_t$

$$\Leftrightarrow X_t + g(t)(\omega_{t+1} - \omega_t)$$

Continuous VE: $X_{t+h} = X_t + g(t)(\omega_{t+h} - \omega_t)$

$$\text{VE-SDE: } dX_t = g(t) d\omega_t$$

SDEs

Discrete VP: $X_{t+1} = \sqrt{1 - \beta_t'} X_t + \sqrt{\beta_t'} \varepsilon_t$

Continuous VP: $X_{t+h} = \sqrt{1 - \beta_t' h} X_t + \sqrt{\beta_t' h} \varepsilon_t$

$$\Leftrightarrow \left(1 - h \frac{\beta_t}{2} + o(h)\right) X_t + \sqrt{\beta_t} (\omega_{t+h} - \omega_t)$$

VP-SDE: $dX_t = -\frac{\beta_t}{2} X_t dt + \sqrt{\beta_t} d\omega_t$

$$P_{X_t|X_0}(x_t|x_0) = \mathcal{N}(x_t | x_0 \sqrt{d_t}, (1-d_t) I)$$

$$d_t = \exp\left(-\int_0^t \beta_s ds\right)$$

Backward process

VP/VE (simple)

Forward: $\begin{cases} dX_t = f(X_t, t)dt + g(t)dW_t \leftarrow \\ X_0 \sim P_{\text{data}} \leftarrow \text{difficult} \end{cases}$

s.t. $P_1 := P_{X_1}$ is simple and known (s.t. $N(0, \sigma^2 I)$).

Goal: define $X_t^{(b)}$: $\begin{cases} dX_t^{(b)} = f^{(b)}(X_t^{(b)}, t)dt + g^{(b)}(t)dW_t \leftarrow \\ X_0^{(b)} \sim P_1 \leftarrow \text{simple} \end{cases}$ difficult

s.t. $q_t := P_{X_t^{(b)}}$ equals $P_{t-t} = P_{X_{t-t}}$.

Backward ODE is easy

Let Y_t satisfy
$$\begin{cases} dY_t = v(Y_t, t)dt \\ Y_0 = y_0 \end{cases}$$

Define $Y_t^{(b)}$ as
$$\begin{cases} dY_t^{(b)} = -v(Y_t^{(b)}, 1-t)dt \\ Y_0^{(b)} = Y_1 \end{cases}$$

Then, $Y_t^{(b)} = Y_{1-t} \quad \forall t.$

Backward SDE scheme

$$\text{SDE} \xrightarrow{?} \text{ODE} \xleftrightarrow{v(\cdot, t) \leftrightarrow -v(\cdot, 1-t)} \text{b-ODE} \xrightarrow{?} \text{b-SDE}$$

Goal: $\begin{cases} dX_t = f(X_t, t)dt + g(t)dW_t \\ X_0 \sim p_0 \end{cases}$

$\begin{cases} dY_t = v(Y_t, t)dt \\ Y_0 \sim p_0 \end{cases}$

save marginals: $P_{X_t} = P_{Y_t}$

How? Study evolution of density.

Evolution of density

- $dX_t = f(X_t, t)dt + g(t)dW_t$

$$X_{t+h} = X_t + h f(X_t, t) + \sqrt{h} g(t) \varepsilon_t$$

- $P_{X_{t+h}} = ? \dots P_{X_t} \dots ?$

1) $X_t \rightarrow \varphi_h(X_t) = X_t + h f(X_t, t)$ — deterministic transform

$$P_{\varphi_h(X_t)}(y) = P_{X_t}(\varphi_h^{-1}(y)) \left| \det \frac{\partial \varphi_h^{-1}}{\partial y} \right| — \text{change of variables}$$

2) $\varphi_h(X_t) \rightarrow X_{t+h} = \varphi_h(X_t) + \sqrt{h} g(t) \varepsilon_t$

$$P_{X_{t+h}}(x) = \int P_{\varphi_h(X_t)}(x-y) P_{\sqrt{h} g(t) \varepsilon_t}(y) dy — \text{convolution}$$

Evolution of density (ODE)

- $X_t \rightarrow \varphi_h(X_t) = X_t + h f(X_t, t)$

$$P_{\varphi_h(X_t)}(y) = P_{X_t}(\varphi_h^{-1}(y)) \left| \det \frac{\partial \varphi_h^{-1}}{\partial y} \right| \Leftrightarrow P_{X_t}(x) = P_{\varphi_h(X_t)}(\varphi_h(x)) \left| \det \frac{\partial \varphi_h}{\partial x} \right|$$

$$X_t \in \mathbb{R} \Rightarrow P_{X_t}(x) = P_{\varphi_h}(x + h f(x, t)) \left(1 + h \frac{\partial}{\partial x} f(x, t) \right) =$$

$$= \left(P_{\varphi_h}(x) + h f(x, t) \frac{\partial}{\partial x} P_{\varphi_h}(x) + \mathcal{O}(h) \right) \left(1 + h \frac{\partial}{\partial x} f(x, t) \right)$$

$$= P_{\varphi_h}(x) + h f(x, t) \frac{\partial}{\partial x} P_{\varphi_h}(x) + h \frac{\partial}{\partial x} f(x, t) \cdot P_{\varphi_h}(x) + \mathcal{O}(h)$$

$$= P_{\varphi_h}(x) + h \frac{\partial}{\partial x} (P_{\varphi_h}(x) f(x, t)) + \mathcal{O}(h)$$

$$\frac{\underset{\downarrow P_{X_t}(x)}{P_{\varphi_h}(x)} - P_{X_t}(x)}{h} = - \frac{\partial}{\partial x} \left(\underset{\downarrow P_{X_t}(x)}{P_{\varphi_h}(x)} f(x, t) \right) + \mathcal{O}(1)$$

Evolution of density (ODE)

- $dX_t = f(X_t, t)dt$

$$\frac{\partial}{\partial t} p_t(x) = - \frac{\partial}{\partial x} (p_t(x) f(x, t))$$

Continuity
equation

Evolution of density (SDE)

- $\varphi_h(X_t) \rightarrow X_{t+h} = \varphi_h(X_t) + \sqrt{h} g(t) \varepsilon_t$

$$\begin{aligned} P_{X_{t+h}}(x) &= \int P_{\varphi_h(X_t)}(x-y) P_{\sqrt{h} g(t) \varepsilon_t}(y) dy = \mathbb{E} P_{\varphi_h}(x - \sqrt{h} g(t) \varepsilon_t) \\ &= \mathbb{E} \left[P_{\varphi_h}(x) - \sqrt{h} g(t) \varepsilon_t \frac{\partial}{\partial x} P_{\varphi_h}(x) + h \frac{g^2(t)}{2} \varepsilon_t^2 \frac{\partial^2}{\partial x^2} P_{\varphi_h}(x) + \bar{o}(h) \right] \\ &= P_{\varphi_h}(x) + 0 + h \frac{g^2(t)}{2} \cdot 1 \cdot \frac{\partial^2}{\partial x^2} P_{\varphi_h}(x) + \bar{o}(h) = \\ &= P_{\varphi_h}(x) + h \frac{g^2(t)}{2} \frac{\partial^2}{\partial x^2} P_{\varphi_h}(x) + \bar{o}(h) \\ &= P_{X_t}(x) - h \frac{\partial}{\partial x} (P_{\varphi_h}(x) f(x, t)) + h \frac{g^2(t)}{2} \frac{\partial^2}{\partial x^2} P_{\varphi_h}(x) + \bar{o}(h) \end{aligned}$$

$$\frac{P_{X_{t+h}}(x) - P_{X_t}(x)}{h} = - \frac{\partial}{\partial x} \left(\underbrace{P_{\varphi_h}(x)}_{\downarrow P_{X_t}(x)} f(x, t) \right) + \frac{g^2(t)}{2} \frac{\partial^2}{\partial x^2} \underbrace{P_{\varphi_h}(x)}_{\downarrow P_{X_t}(x)} + \bar{o}(1)$$

Evolution of density

- $dX_t = f(X_t, t)dt$

$$\frac{\partial}{\partial t} p_t(x) = - \frac{\partial}{\partial x} (p_t(x) f(x, t))$$

Continuity
equation

- $dX_t = f(X_t, t)dt + g(t)dW_t$

$$\frac{\partial}{\partial t} p_t(x) = - \frac{\partial}{\partial x} (p_t(x) f(x, t)) + \frac{g^2(t)}{2} \frac{\partial^2}{\partial x^2} p_t(x)$$

Fokker-Planck (-Kolmogorov) equation

Interpretation
Conservation laws (thanks to S.V. Shaposhnikov
and T.I. Krasovitsky
for their PDE course)

Let $u(x, t)$ be density of a substance at time t .
Amount of substance in $[x_0, x_0 + \Delta x]$ is

$$\int_{x_0}^{x_0 + \Delta x} u(x, t) dx.$$

Change from time t_0 to $t_0 + \Delta t$ is

$$\int_{x_0}^{x_0 + \Delta x} u(x, t_0 + \Delta t) dx - \int_{x_0}^{x_0 + \Delta x} u(x, t_0) dx.$$

Interpretation

Let $F(x, t)$ be flow of the substance, i.e. the amount of flowed substance in the point x_0 from t_0 to $t_0 + \Delta t$ is

$$\int_{t_0}^{t_0 + \Delta t} F(x, t) dt.$$

Then,

$$\int_{x_0}^{x_0 + \Delta x} u(x, t_0 + \Delta t) - u(x, t_0) dx = \int_{t_0}^{t_0 + \Delta t} F(x_0, t) - F(x_0 + \Delta x, t) dt$$

$$\int_{x_0}^{x_0 + \Delta x} \int_{t_0}^{t_0 + \Delta t} \frac{\partial}{\partial t} u(x, t) dx dt = - \int_{x_0}^{x_0 + \Delta x} \int_{t_0}^{t_0 + \Delta t} \frac{\partial}{\partial x} F(x, t) dx dt$$

Conservation laws

$$\frac{\partial}{\partial t} u(x, t) = - \frac{\partial}{\partial x} F(x, t)$$

General FPE:

$$\frac{\partial}{\partial t} p_t(x) = - \frac{\partial}{\partial x} (p_t(x) f(x, t)) + \frac{g^2(t)}{2} \frac{\partial^2}{\partial x^2} p_t(x)$$

- $dX_t = f(X_t, t)dt$ $F(x, t) = f(x, t) \cdot p_t(x)$
- $dX_t = g(t)dW_t$ $F(x, t) = - \frac{g^2(t)}{2} \frac{\partial}{\partial x} p_t(x)$
- $dX_t = f(X_t, t)dt + g(t)dW_t$ $F(x, t) = f(x, t) \cdot p_t(x) - \frac{g^2(t)}{2} \frac{\partial}{\partial x} p_t(x)$

Uniqueness

Thm Under regularity conditions, FPE

$$\begin{cases} \frac{\partial}{\partial t} p_t(x) = - \frac{\partial}{\partial x} (p_t(x) f(x, t)) + \frac{g^2(t)}{2} \frac{\partial^2}{\partial x^2} p_t(x) \\ p_0(x) = \hat{p}(x) \end{cases}$$

has unique solution, i.e. $\Leftrightarrow$ corresponds to evolution of density of SDE

$$\begin{cases} dX_t = f(X_t, t)dt + g(t)dW_t \\ X_0 \sim \hat{p} \end{cases}$$

SDE $\rightarrow$ ODE

$$dX_t = f(X_t, t)dt + g(t)dW_t$$

$$\frac{\partial}{\partial t} p_t(x) = - \frac{\partial}{\partial x} (p_t(x) f(x, t)) + \frac{g^2(t)}{2} \frac{\partial^2}{\partial x^2} p_t(x)$$

$$= - \frac{\partial}{\partial x} \left(p_t(x) f(x, t) - \frac{g^2(t)}{2} \frac{\partial}{\partial x} p_t(x) \right)$$

$$= - \frac{\partial}{\partial x} \left(p_t(x) \left[f(x, t) - \frac{g^2(t)}{2} \frac{\partial}{\partial x} \log p_t(x) \right] \right)$$

$$dY_t = \left(f(Y_t, t) - \frac{g^2(t)}{2} \frac{\partial}{\partial x} \log p_t(Y_t) \right) dt$$

— corresponding forward ODE.

ODE₂ b-ODE

$$dY_t = v(Y_t, t)dt, \quad v(x, t) = f(x, t) - \frac{g^2(t)}{2} \frac{\partial}{\partial x} \log P_t(x)$$

↓

$$dY_t^{(b)} = -v(Y_t^{(b)}, t-t)dt = \left(-f(Y_t^{(b)}, t-t) + \frac{g^2(t)}{2} \frac{\partial}{\partial x} \log P_{t-t}(Y_t^{(b)}) \right) dt$$

– backward ODE

b-ODE 2 b-SDE

$$dY_t^{(b)} = \underline{-v(Y_t^{(b)}, 1-t)} dt \quad (q_t := P_{Y_t^{(b)}} = P_{Y_{1-t}} = P_{1-t})$$

$$\downarrow$$
$$\frac{\partial}{\partial t} q_t(x) = - \frac{\partial}{\partial x} (q_t(x) \underline{-v(x, 1-t)})$$

$$= - \frac{\partial}{\partial x} \left(q_t(x) \left(\underline{-v(x, 1-t) + \frac{g^2(1-t)}{2} \frac{\partial}{\partial x} \log q_t(x)} \right) \right) +$$
$$+ \frac{g^2(1-t)}{2} \frac{\partial^2}{\partial x^2} q_t(x)$$

$$\uparrow$$
$$dX_t^{(b)} = \left(\underline{-v(X_t^{(b)}, 1-t) + \frac{g^2(1-t)}{2} \frac{\partial}{\partial x} \log P_{1-t}(X_t^{(b)})} \right) dt$$
$$+ g(1-t) dW_t$$

b-ODE 2 b-SDE

$$dX_t^{(b)} = \left(-v(X_t^{(b)}, 1-t) + \frac{g^2(1-t)}{2} \frac{\partial}{\partial x} \log p_{1-t}(X_t^{(b)}) \right) dt + g(1-t) dW_t$$

Recall $v(x, t) = f(x, t) - \frac{g^2(t)}{2} \frac{\partial}{\partial x} \log p_t(x)$

$$dX_t^{(b)} = \left(-f(X_t^{(b)}, 1-t) + g^2(1-t) \frac{\partial}{\partial x} \log p_{1-t}(X_t^{(b)}) \right) dt + g(1-t) dW_t$$

Summary

$$dX_t = f(X_t, t)dt + g(t)dW_t$$

↓

$$dY_t = \left(f(Y_t, t) - \frac{g^2(t)}{2} \frac{\partial}{\partial x} \log p_t(Y_t) \right) dt$$

↓

$$dY_t^{(b)} = \left(-f(Y_t^{(b)}, 1-t) + \frac{g^2(1-t)}{2} \frac{\partial}{\partial x} \log p_{1-t}(Y_t^{(b)}) \right) dt$$

↓

$$dX_t^{(b)} = \left(-f(X_t^{(b)}, 1-t) + g^2(1-t) \frac{\partial}{\partial x} \log p_{1-t}(X_t^{(b)}) \right) dt + g(1-t)dW_t$$

VE-SDE

$$dX_t = g(t) dW_t$$

$$dY_t = -\frac{g^2(t)}{2} \frac{\partial}{\partial x} \log P_t(Y_t) dt$$

$$dY_t^{(b)} = \frac{g^2(1-t)}{2} \frac{\partial}{\partial x} \log P_{1-t}(Y_t^{(b)}) dt$$

$$dX_t^{(b)} = g^2(1-t) \frac{\partial}{\partial x} \log P_{1-t}(Y_t^{(b)}) dt + g(1-t) dW_t$$

Sampling

Both b-ODE $Y_t^{(b)}$ and b-SDE $X_t^{(b)}$ generate the reversed dynamics P_{1-t} , given $X_0^{(b)} \sim P_1$.
($Y_0^{(b)} \sim P_1$)

$$dY_t^{(b)} = \left(-f(Y_t^{(b)}, 1-t) + \frac{g^2(t)}{2} \frac{\partial}{\partial x} \log P_{1-t}(Y_t^{(b)}) \right) dt$$

$$dX_t^{(b)} = \left(-f(X_t^{(b)}, 1-t) + g^2(1-t) \frac{\partial}{\partial x} \log P_{1-t}(X_t^{(b)}) \right) dt + g(1-t) dW_t$$

Need to know $\frac{\partial}{\partial x} \log P_t(x) \leftarrow$ learn by neural net.

Summary

- Fokker-Planck equation

$$\frac{\partial}{\partial t} p_t(x) = - \frac{\partial}{\partial x} (p_t(x) f(x, t)) + \frac{g^2(t)}{2} \frac{\partial^2}{\partial x^2} p_t(x)$$

- SDE $\rightarrow$ ODE $\xrightarrow{v \rightarrow -v}$ ODE $\rightarrow$ SDE
FPE $\rightarrow$ CE $\quad$ CE $\rightarrow$ FPE

- $dX_t^{(b)} = \left(-f(X_t^{(b)}, 1-t) + g^2(1-t) \frac{\partial}{\partial x} \log p_{1-t}(X_t^{(b)}) \right) dt + g(1-t) dW_t$

Denoising Score Matching

Loss: $X_0 \sim p_{\text{data}}$, $\varepsilon \sim \mathcal{N}(0, I)$, $X_t = \alpha_t X_0 + \sigma_t \varepsilon$

$$\int_0^1 \mathbb{E} \left\| S_\theta(X_t, t) - \frac{\partial}{\partial x} \log p_t(X_t | X_0) \right\|^2 dt \rightarrow \min_{\theta}$$

- $p_{t|0}(x_t | x_0) = \mathcal{N}(x_t | \alpha_t x_0, \sigma_t^2 I) = C \cdot \exp\left(-\frac{1}{2\sigma_t^2} \|x_t - \alpha_t x_0\|^2\right)$

$$\frac{\partial}{\partial x_t} \log p_{t|0}(x_t | x_0) = -\frac{1}{\sigma_t^2} (x_t - \alpha_t x_0) = -\frac{\varepsilon}{\sigma_t}$$

- Parameterize $S_\theta(x_t, t) = \frac{-\varepsilon_\theta(x_t, t)}{\sigma_t}$

Loss: $\int_0^1 \omega_t \mathbb{E} \left\| \varepsilon_\theta(x_t, t) - \varepsilon \right\|^2 dt \rightarrow \min_{\theta}$

Sampling (VE)

$$dX_t^{(b)} = g^2(1-t) S_{\theta}(X_t^{(b)}, 1-t) dt + g(1-t) dW_t$$

$$X_T \sim \mathcal{N}(0, \sigma_T^2)$$

$$\begin{aligned} X_{t-h} &\leftarrow X_t + h g^2(t) S_{\theta}(X_t, t) + \sqrt{h} g(t) \varepsilon_t \\ &= X_t - h g^2(t) \frac{\varepsilon_{\theta}(X_t, t)}{\sigma_t} + \sqrt{h} g(t) \varepsilon_t \end{aligned}$$

generalizes DDPM sampling

ODE sampling (VE)

$$dY_t^{(b)} = \frac{g^2(1-t)}{2} S_\theta(Y_t^{(b)}, 1-t) dt$$

$$\begin{aligned} Y_{t-h} &\leftarrow Y_t + h \cdot \frac{g^2(t)}{2} S_\theta(Y_t, t) = \\ &= Y_t - h \frac{g^2(t)}{2} \frac{\mathcal{E}_\theta(Y_t, t)}{G_t} \end{aligned}$$

Example: $G_t^2 = t^2$ $g(t) = \sqrt{2t}$

$$Y_{t-h} \leftarrow Y_t - h \mathcal{E}_\theta(Y_t, t)$$

Summary

- $\int_0^t \omega_t \mathbb{E} \left\| s_\theta(X_t, t) - \frac{\partial}{\partial x} \log p_{t|0}(X_t | X_0) \right\|^2 dt \rightarrow \min_{\theta}$
 $\Leftrightarrow$

$$\int_0^t \omega_t \mathbb{E} \left\| \varepsilon_\theta(X_t, t) - \varepsilon \right\|^2 dt \rightarrow \min_{\theta}$$

- SDE sampling $\Leftrightarrow$ DDPM

- ODE sampling : simple, interpretable

